

Aniruddha Sen | Curriculum Vitae

– Austin, TX

☎ (413) 275-3778 • ✉ aniruddhasen@utexas.edu • 📄 aniruddhasen.github.io

Education

The University of Texas at Austin

PhD in Computer Science

2024–

Advised by Scott Aaronson and Nick Hunter-Jones. Specializing in quantum computing theory. Supported by a TQI Graduate Fellowship.

The University of Massachusetts Amherst

Bachelor of Science in Computer Science

2020–2024

Graduated *Summa Cum Laude*, *Departmental Honors with Greatest Distinction*. Double Major in Mathematics.

Research

1. *Nearly Time-Optimal Pure State Tomography with Pauli Measurements*. S. Grewal, M. Gupta, W. He, **A. Sen**, M. Singhal. *to appear in Foundations of Computer Science (FOCS)*, 2026. arXiv:2601.04444.
2. *Learning Sparse Quantum States*. **A. Sen**. arXiv:2609.12219 (2026).
3. *Local Random Quantum Circuits Converge to the Porter-Thomas Distribution in Polynomial Depth*. **A. Sen**, N. Hunter-Jones. arXiv:2610.02125 (2026).
4. *Device-Independent Conference Keys from Parity-Extended Games*. S. Chakraborty, R. Ramachandran, **A. Sen**. arXiv:2610.01025 (2026).
5. *Classical Shadows with Arbitrary Group Representations*. M. West, F. Sauvage, **A. Sen**, R. Forestano, D. Wierichs, N. Killoran, D. Grinko, M. Cerezo, M. Larocca. *Quantum Techniques in Machine Learning (QTML)*, 2025. arXiv:2604.01429.
6. *Learning Superpositions of Product States*. **A. Sen**. *Theory of Quantum Computation, Communication and Cryptography (TQC) 2025* (poster).
7. *Multipartite Entanglement Distribution in Quantum Networks via Subgraph Complementations*. **A. Sen**, K. Goodenough, D. Towsley. *Quantum* **9**, 1911 (2025). arXiv:2308.13700.
8. *Quantum Circuit Simulation via ZX-Calculus and Tensor Network Contraction*. **A. Sen**, Y. Pang, E. Solomonik. *Workshop on Sparse Tensor Computations*, 2023 (poster).
9. *Diverse Community Data for Benchmarking Data Privacy Algorithms*. **A. Sen**, C. Task, D. Kapur, G. Howarth, K. Bhagat. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2306.13216.

Experience

Visa Research

Foster City, CA

Quantum Research Intern

Summer 2026

Mentored by: Dr Suvradip Chakraborty

- Worked on extending device-independent quantum key distribution with 2 players to N -player device-independent conference key agreement by defining and studying 'parity-extended' games, enabling new rigorous proofs of security

Theoretical Division, Los Alamos National Laboratory

Los Alamos, NM

Research Fellow

Summer 2025

Mentored by: Dr Martin Larocca

- Conducted research in quantum learning theory and quantum algorithms as part of the LANL Quantum Computing Summer School Fellowship.
- Utilized techniques from representation theory and analysis to show exponential separations in sample complexity in the setting of shadow tomography for learning group invariant properties with conjugate access, with applications such as estimating entanglement monotones and novel approaches for the hidden subgroup problem.

LPNA, The University of Illinois Urbana-Champaign

Quantum Research Intern

Advised by: Professor Edgar Solomonik

- Designed and analyzed algorithms for the simulation of Clifford quantum circuits and general quantum circuits by using ZX-calculus and tensor network contractions combined in a hybrid approach.
- Compared Clifford simulation algorithm to measurement based quantum computation using gFlow measure
- Parameterized general circuit simulation complexity through a novel parameter – minimum rank of its adjacency matrix

Champaign, IL

Summer 2023

ACQuIRE Lab, UMass Amherst

Undergraduate Researcher

Advised by: Distinguished Professor Don Towsley

- Designed and analyzed algorithms for generating multipartite entanglement in a quantum network, and also evaluated their performance through classical simulation.
- Obtained a quadratic improvement over prior work on distributing an arbitrary graph state across a quantum network

Amherst, MA

2023–2024

Quantum Information Theory Lab, UMass Amherst

Undergraduate Research Assistant

Advised by: Professor Chen Wang

- Worked with superconducting *fluxonium* qubits and studied circuit quantum electrodynamics (cQED) to carry out 9 experiments with qubit controllers using PYNQ software
- Executed resonator spectroscopy and pulse programming techniques to calibrate ZCU216 Xilinx FPGA boards

Amherst, MA

2022-2023

Invited Talks

- **Pauli Measurements Are Near-Optimal for Pure State Tomography.** March 2026
IQC Math and CS seminar, University of Waterloo
- **Nearly Time-Optimal Pure State Tomography with Pauli Measurements.** June 2026
Quantum seminar, Visa Research

Workshops

Cornell, Maryland, Max Planck Pre-doctoral Research School (CMMRS) MPI **Saarbrücken, Germany**
2023

- Attended research seminars at CMMRS 2023 as part of a fully-funded invitation to the Max Planck institute (MPI)

Awards and Recognitions

2025: UT Austin, TQI Graduate Fellowship – Selected as one of four recipients of the Texas Quantum Institute (TQI) Graduate Fellowship in 2025 which provides support for two years of graduate work

2024: Yale, YQuantum Classiq Winner – Awarded First place for the Classiq challenge at the YQuantum hackathon for implementation of a sparse quantum state preparation algorithm

2023: IBM, Qiskit Advocate – Contributing to quantum computing research and applications as part of a select global community recognized by IBM

2023: MIT iQuHACK, Grand Prize Winner – Awarded First place for the IonQ challenge at MIT's annual quantum computing hackathon

2022: ICPC (International Collegiate Programming Contest) – Ranked First in the Northeast North America Regional Contest, New England (Outside Boston)

2020: UMass, Chancellor's Award – Received a \$64000 tuition waver on the basis of academic performance

2020-2024: UMass CICS, Dean's List – All semesters

Leadership and Extra-Curricular

Gave a talk and mentored a team of 10 students at the NYU Abu Dhabi International Quantum Computing Hackathon & Conference (May 2025). Teaching Assistant for CS 395T / Phy 394U – *Quantum Information Theory* at UT Austin (Fall 2026). Club tennis player and chess enthusiast.